MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 7403

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, pulsed beam power

Outline, Base, Envelope: See figure 1

Cathode --- Coated unipotential

Base connections:

Pin No. ---	1	2	4	6	7	8	Cap
Element ---	h	h	g2	g1	nc	k,g3	a

ABSOLUTE RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ehk	Rk	Ik	Ic1
Unit:	V	Vdc	Vdc	Vdc	v	Ohms	mAdc	mAdc
Maximum:	6.9	4,000	0	850	500	----	150	+5.0
Minimum:	5.7	---	-200	---	---	----	---	----
TEST CONDITIONS:	6.3	600	0	300	0	825	---	----

ABSOLUTE RATINGS:

Parameter:	Pp	Pg2	egk	ik	tk	TE	Alt
Unit:	W	W	v	a	sec	°C	ft
Maximum:	40	3.5	+220	10	--	300	830,000
Minimum:	--	---	----	(see note 6)	30	(see note 11)	(see note 7)
TEST CONDITIONS:	--	---	----	---	---	---	---

GENERAL:

Qualification - Required

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:38Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Quality conformance inspection, part 1							
1301	Heater current		0.65	II	If	1.55	1.85	A
1336	Heater-cathode leakage	Ehk = +450 Vdc Ehk = -450 Vdc	0.65	II	Ihk	----	50	μAdc
1266	Total grid current	t = 120 See note 8	0.65	II	Ic1	0	-5.0	μAdc
1256	Anode current (1)	See note 8	0.65	II	Ib	28	37	mAdc
1256	Screen current	See note 8	0.65	II	Ic2	----	3.0	mAdc
1306	Transconductance	See note 8	0.65	II	Sm	5,000	7,000	μmhos
1256	Anode current (2)	Ebb = 4,000 Vdc; Ec2 = 800 Vdc; Ec1 = -180 Vdc; Rp = 2.0 Meg See note 10	0.65	II	Ib	0	300	μAdc
----	Peak anode current(1)	See notes 1 and 10	0.65	II	ib	6.0	---	a
----	Peak screen current	See notes 1 and 10	0.65	II	ic2	0	3.0	a
----	Peak grid current	See notes 1 and 10	0.65	II	ic1	----	2.0	a
1201	Short and discontinuity detection		0.4	II	---	----	----	---
	Quality conformance inspection, part 2							
1041	Shock	450 G; Ehk = + 450 Vdc; Ec1 = -10 Vdc; Rp = 2,000 ohms; Eb = 250 Vdc; Ec2 = 100 Vdc	15	S2	---	----	----	---
----	Vibration fatigue	5G; F = 50 to 500 to 50 Hz; Ef = 6.3 V; no other voltage applied See note 9	6.5	See note 2	---	----	----	---
----	Post-shock and vibration-fatigue end points:							
1336	Heater-cathode leakage	Ehk = +450 Vdc Ehk = -450 Vdc	---- ----	--- ---	Ihk Ihk	---- ----	100 100	μAdc μAdc
----	Peak anode current (1)		----	---	ib	5.5	---	a
1331	Direct-interelectrode capacitance	No shield	6.5	S3	Cgp Cin Cout	--- 13 7.5	2.2 21 12.5	pF pF pF
----	Peak anode current(2)	Peak anode current (1) conditions except, Ef = 5.7 Vac	2.5	I	ib Δ ib Ef	5.8 ----	--- 10	a %
1101	Secureness of base, base insert and cap		6.5	S3	---	----	----	---
1111	Base pin solder depth (rigid leads)		6.5	S3	---	----	----	---
1211	Insulation of electrodes		4.0	S3	---	10	----	Meg
1105	Permanence of marking		----	---	---	----	----	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:38Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Quality conformance inspection, part 3							
1506	Heater-cycling life test	Ef = 7.0 V; Ehk = 300 Vac; Ec1 = Ec2 = Eb = 0; 1 minute on, 4 minutes off See note 3	---	---	---	---	---	---
----	Heater-cycling life-test end points:							
1336	Heater-cathode leakage	Ehk = + 450 Vdc	---	---	Ihk	---	150	μAdc
		Ehk = - 450 Vdc	---	---	Ihk	---	150	μAdc
1301	Heater current		---	---	If	1.5	1.9	A
----	Pulse stability life test (5 hours)	Ef = 6.0 Vac; Ec1 = -180 Vdc; Ec2 = 800 Vdc; Eb = 600 Vdc; egk = + 150V; tp = 4 μs; duty factor = 0.001	4.0	F	---	---	---	---
----	Pulse stability life-test end points: (5 hours)							
----	Peak anode current (1)		---	---	ib	5.8	---	a
----	Peak anode current (1)	Except Ef = 6.0 Vac	---	---	Δib Ef	---	10	%
----	Pulse life test (500 hours)	Pulse stability life-test conditions except, Ef = 6.3 Vac	---	Group B	---	---	---	---
----	Pulse life test (5,000 hours)	See note 4	---	---	---	---	---	---
----	Pulse life-test end points: (500 hours)							
----	Inoperatives	See note 5	---	---	---	---	---	---
----	Peak anode current (1)		---	---	ib	5.0	---	a

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:38Z
Check the source to verify that this is the current version before use.

NOTES:

1. After minimum preheat of 60 seconds using Ef = 6.3 V only, the tube shall be tested in the circuit of figure 2 under the following pulse conditions: Use rectangular wave modulation on the grid, pulse width to be approximately 4 microseconds, repetition rate = approximately 250 pulses per second, pulse amplitude to be 330 volts peak. Apply anode and screen voltages simultaneously. Time of test = 5 minutes maximum.

2. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6-months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

3. The no-load to steady-state full-load regulation of the heater voltage shall be not more than 3.0 percent. This test shall be made on a lot-by-lot basis.

4. For the purpose of data collection to substantiate life expectancy, 20 percent of the tubes placed on pulse life for the 500-hour test shall be continued on life until failure or 5,000 hours have elapsed.

5. An inoperative, as referenced in life test, is defined as a tube having one or more of the following defects: discontinuity, permanent shorts, and air leaks. (See method 1201.)

6. See rating chart, figure 3, duty factor (10,000 microsecond averaging time) is defined as the ratio of ON time in microseconds to 10,000 microseconds. ON time is defined as the sum of the duration of all individual pulses that occur during any 10,000 microsecond interval.

7. See altitude rating chart, figure 4.

8. All potentials are referred to negative side of cathode resistor.

9. Vibration fatigue test.

   (a) This test shall be performed with an average acceleration level equal to the specified value while being swept through the frequency range of 50 to 500 to 50 hertz.

   (b) The tubes shall be mounted in each of three planes, X, Y, and Z for one-third of the total excitation time.

   (c) The frequency in hertz shall be scanned continuously up and down through the entire frequency range at the rate of approximately 2 minutes.

   (d) The total excitation time for the test shall be 96 hours.

10. Any tubes classed as catastrophic failures due to arcing during test or any tubes that arc sufficiently during test to indicate a continued arcing difficulty shall be considered a defective and shall be included in the AQL for the combined defectives for attributes in quality conformance inspection test, part 1.

11. Cooling shall be employed if the additive effects of simultaneous operation at high altitude and high dissipation or proximity to other heat sources in the equipment tend to cause the maximum bulb temperature to be exceeded.

Custodian:
Air Force - 80

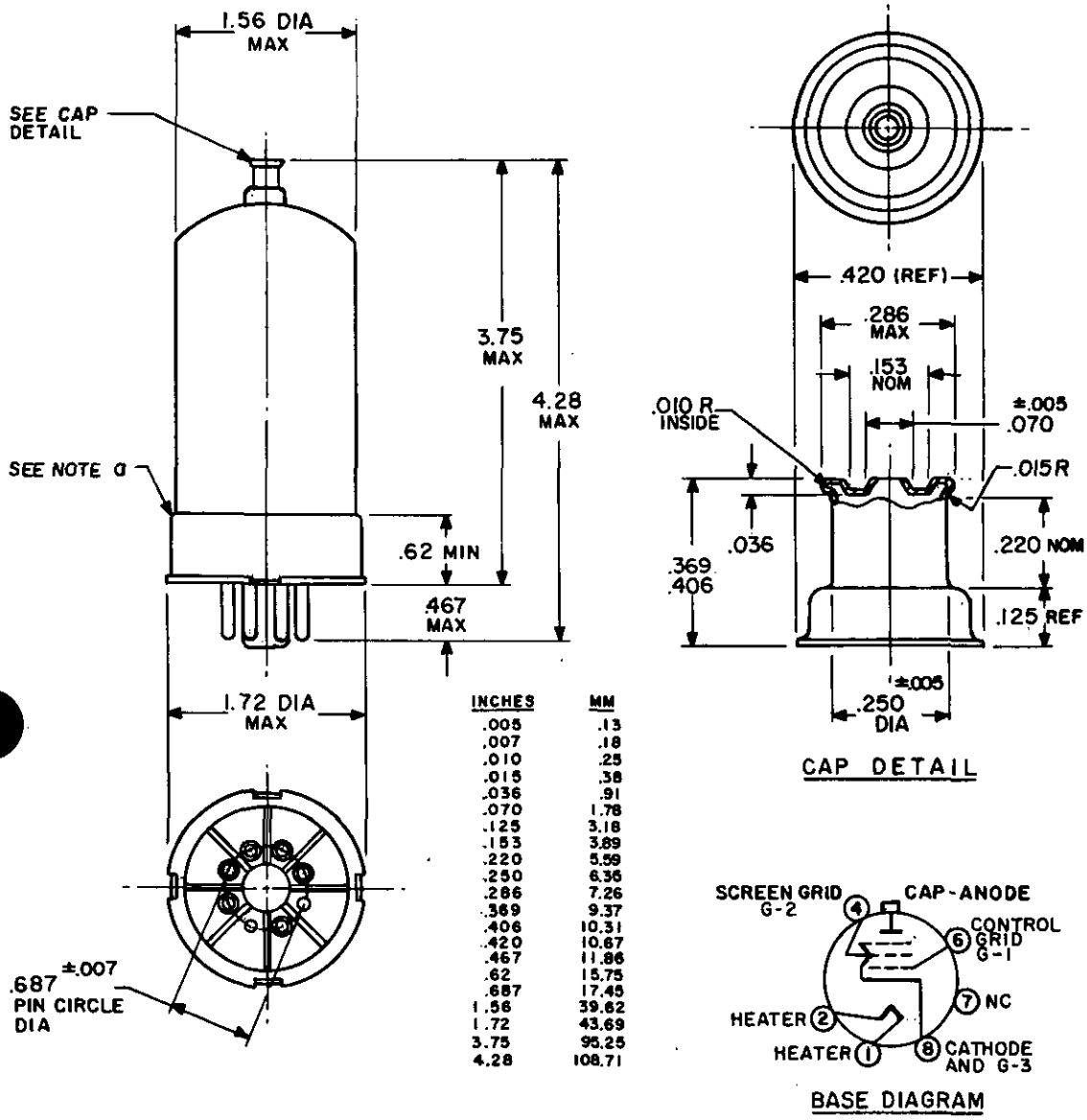
Preparing activity:
Air Force - 80

(Project 5960-906)

Review activities:
Air Force - 11, 17

User activities:
Air Force - 17, 80

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:38Z
Check the source to verify that this is the current version before use.



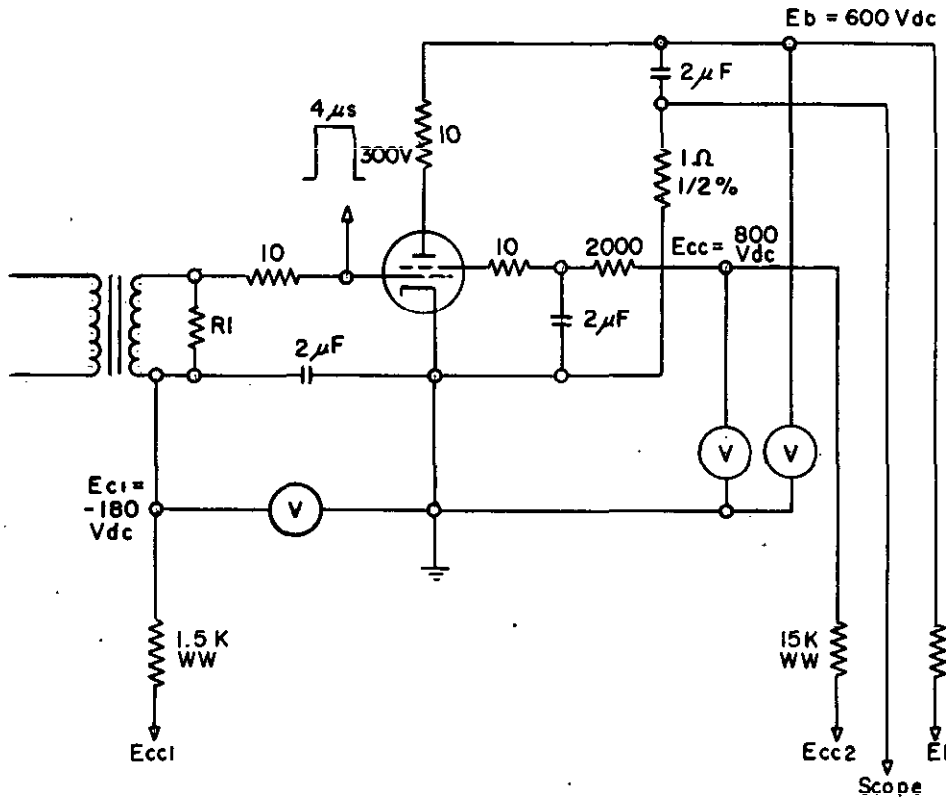
INCHES	MM
.005	.13
.007	.18
.010	.25
.015	.38
.036	.91
.070	1.78
.125	3.18
.153	3.89
.220	5.59
.250	6.36
.286	7.26
.369	9.37
.406	10.31
.420	10.67
.467	11.86
.62	15.75
.687	17.45
1.56	39.62
1.72	43.69
3.75	95.25
4.28	108.71

NOTES:

a. Large wafer octal with metal sleeve.
b. Dimension are in inches.
c. Unless otherwise specified, tolerance is ±.005

FIGURE 1. Outline drawing of electron tube type 7403

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:38Z
Check the source to verify that this is the current version before use.



NOTES:

a. Adjust R1 for minimum ringing and overshoot.
b. Pulse repetition rate = 250 pulses per second. Duty cycle .001.

FIGURE 2 Test circuit

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:38Z
Check the source to verify that this is the current version before use.

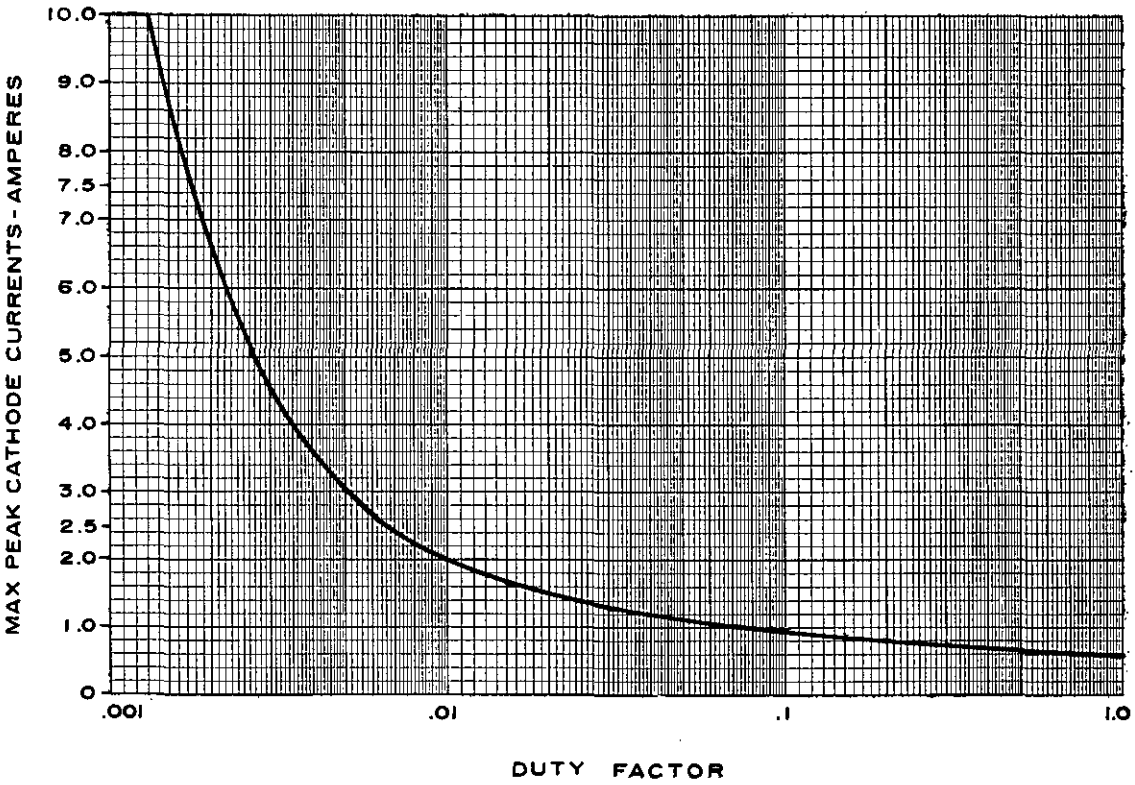


FIGURE 3 Rating chart

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:38Z
Check the source to verify that this is the current version before use.

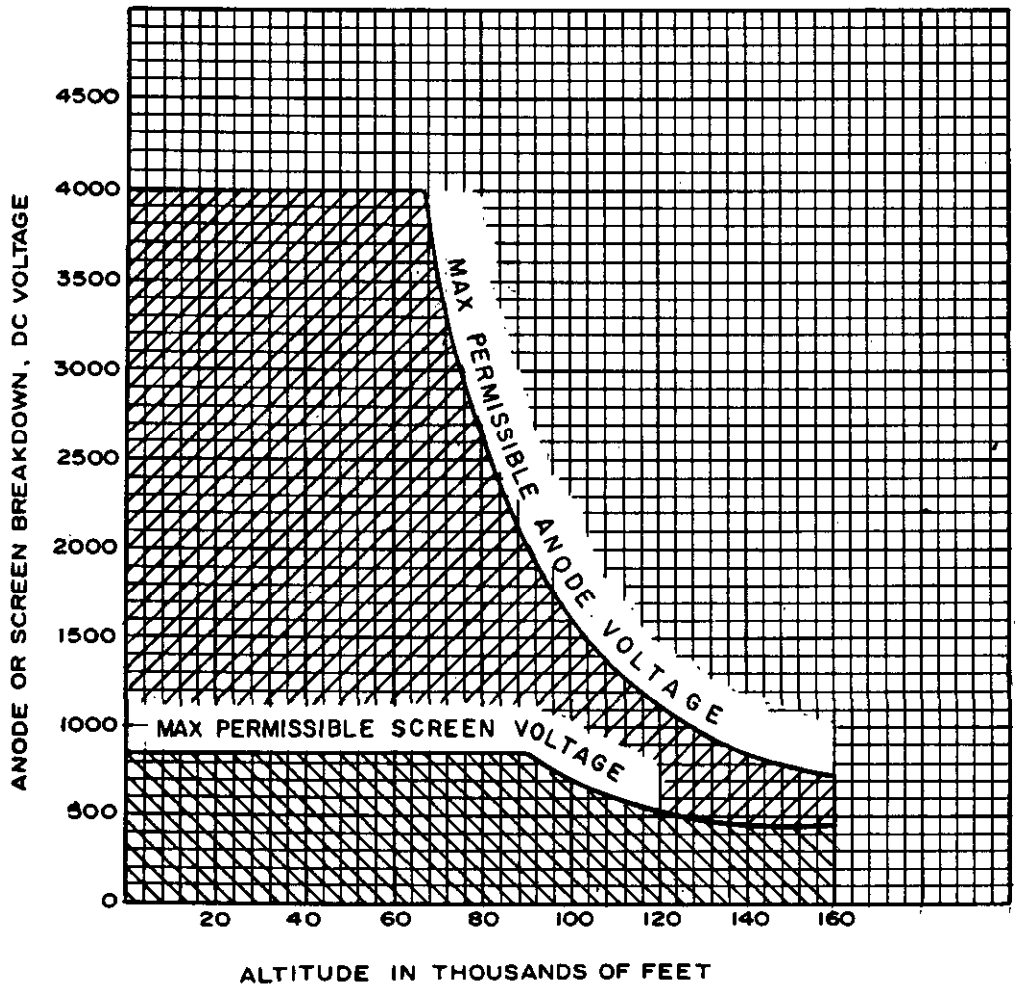


FIGURE 4. Altitude rating chart

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:38Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

C16061

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:38Z
Check the source to verify that this is the current version before use.

FOLD

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF THE AIR FORCE
DoD-318

U.S.MAIL

SMAMA/MMREE
Wright-Patterson AFB OH 45433

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:38Z
Check the source to verify that this is the current version before use.